Where airlines lose time: carrier-attributable delay in US domestic flights

This analysis uses US DOT (BTS) flight data rather than Singapore data, since CAAS/data.gov.sg only publishes aggregate monthly figures for Changi, and no flight-level, cause-coded delay data is publicly available. BTS was chosen as the best available public dataset with the granularity needed for this kind of analysis.

Question. Which airports or carriers show disproportionately high carrier-attributable delay, and does time-of-day or day-of-week make it worse?

What "carrier-attributable" means. BTS splits each delay of 15+ minutes into five causes. *Carrier* delay covers causes inside the airline's control: maintenance, crew, cleaning, fuelling, baggage. BTS has no separate "mechanical" field, so this is **not** a measure of mechanical problems. Weather, air-traffic-system (NAS) and security delays sit outside the airline's control. Late-arriving-aircraft delay is knock-on delay from an earlier flight and is kept separate.

Data: Jan 2024 – Dec 2025, 13,844,730 completed flights. Carrier-attributable delay averages 5.2 min per flight and is 33% of all cause-coded delay minutes.

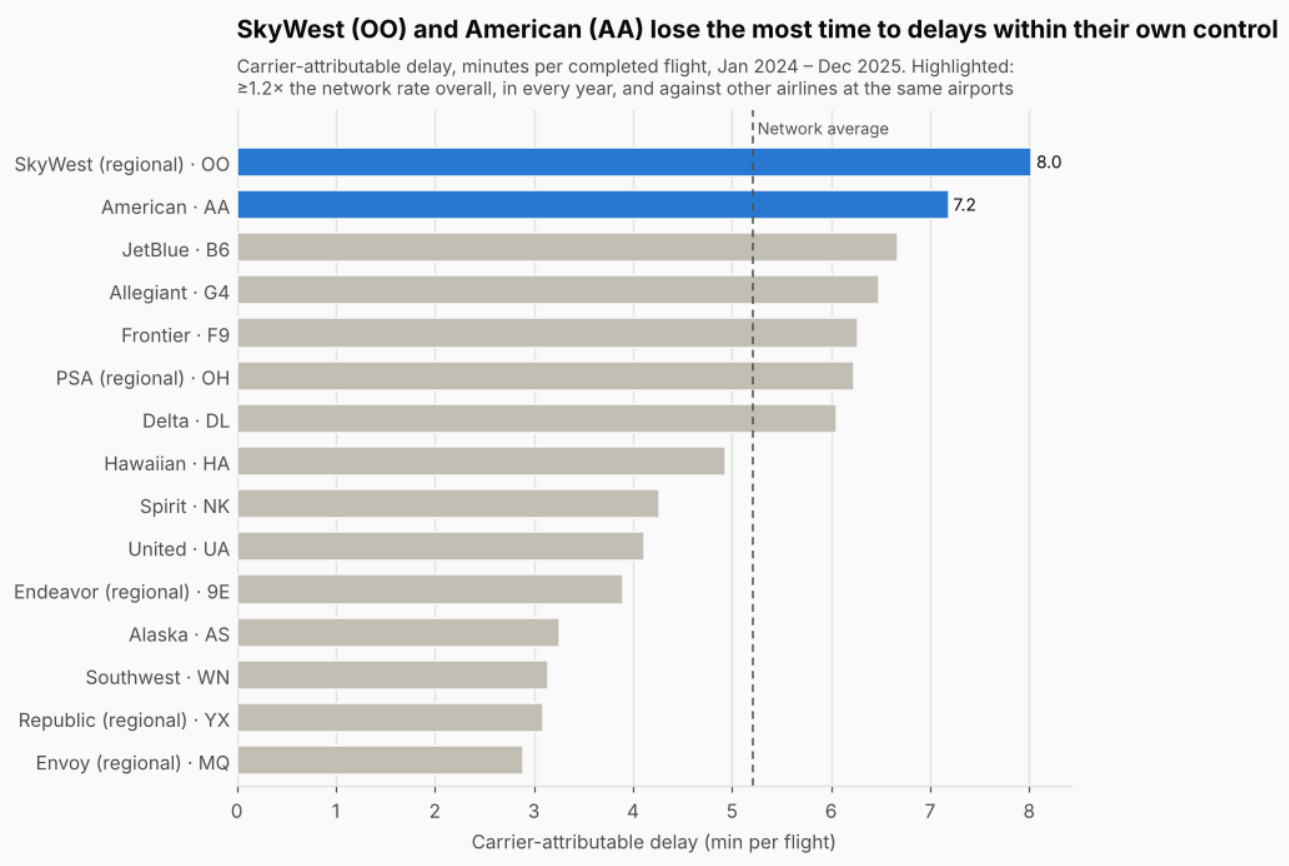
Summary

SkyWest (OO) and American (AA) have disproportionately high carrier-attributable delay: 8.0 and 7.2 minutes per flight, against a network average of 5.2. They stay at least 1.3× the rate of other airlines flying from the same airports, in 2024 and 2025 separately.

Time of day makes it worse. Across the network, carrier-attributable delay per flight is 2.0× higher at 22:00 than at 10:00, and Sunday is the worst day of the week. But for both flagged carriers the biggest excess is in the **first wave**: departures scheduled before 09:00.

Recommendation: SkyWest (OO) and American (AA) should target first-wave departure readiness. Bringing their pre-09:00 departures down to the network's pre-09:00 rate would remove about **20,600 carrier-attributable delay hours a year**. That is before counting the knock-on delay those early flights pass to the rest of the day.

1 · Carriers



Finding 1 · Carriers. Two airlines meet every test:

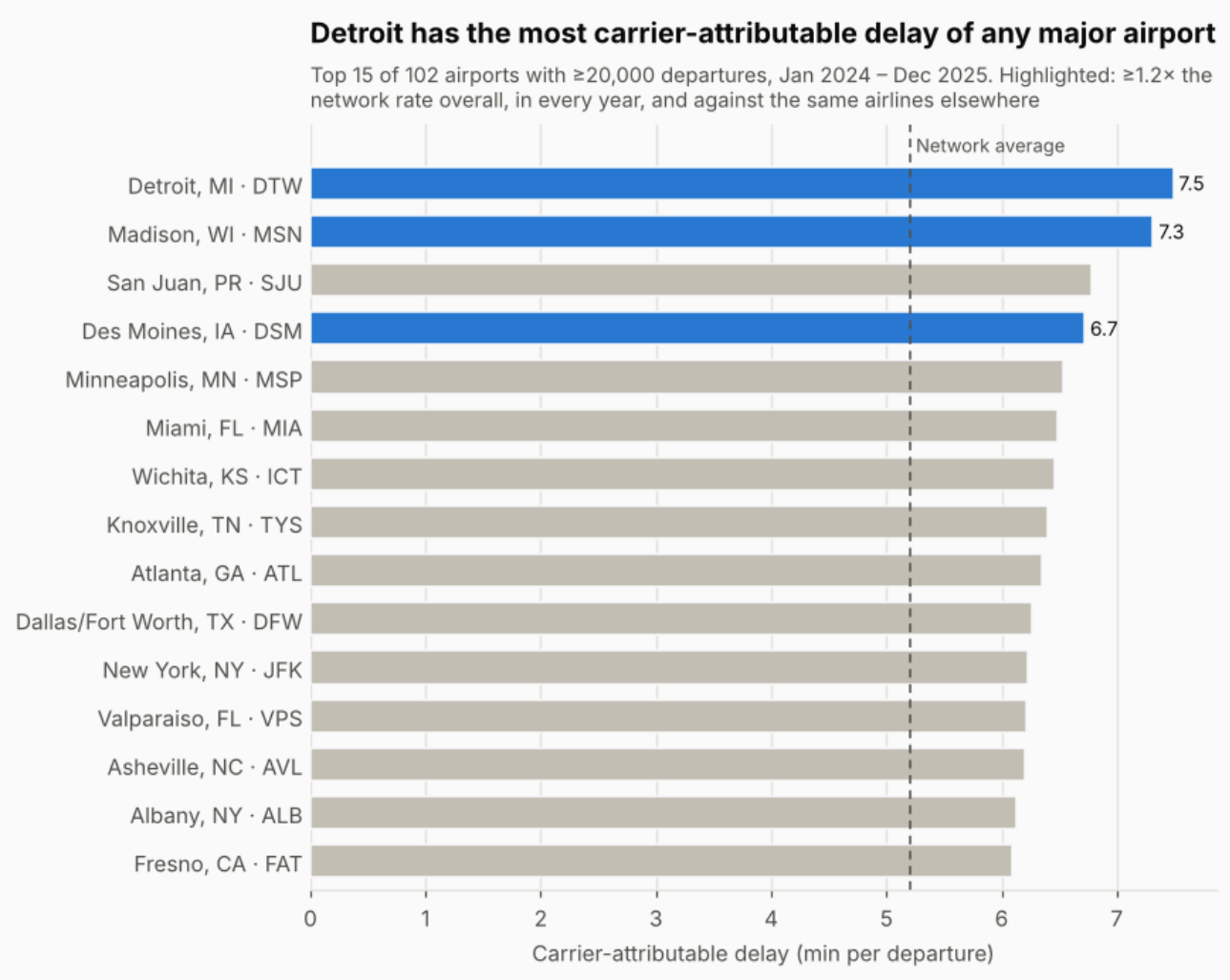
- **SkyWest (OO)**: 8.0 min per flight, 1.54× the network; 1.34× after controlling for airports flown; 2024: 1.51×, 2025: 1.56×. 36,500 excess delay hours a year.
- **American (AA)**: 7.2 min per flight, 1.38× the network; 1.33× after controlling for airports flown; 2024: 1.43×, 2025: 1.32×. 31,600 excess delay hours a year.

They fail in different ways. When a carrier-caused delay happens, the typical one lasts 35 min at SkyWest and 17 min at American (network: 19 min). They also happen on 11% and 15% of flights (network: 12%). So SkyWest's delays are longer, and American's are more frequent.

Above 1.2× overall but not in every year or after the mix control: JetBlue (B6), Allegiant (G4) and Frontier (F9). For example, JetBlue's rate fell from 1.54× in 2024 to 1.02× in 2025.

2 · Airports

Grouped by **origin** airport, where the delay was incurred.



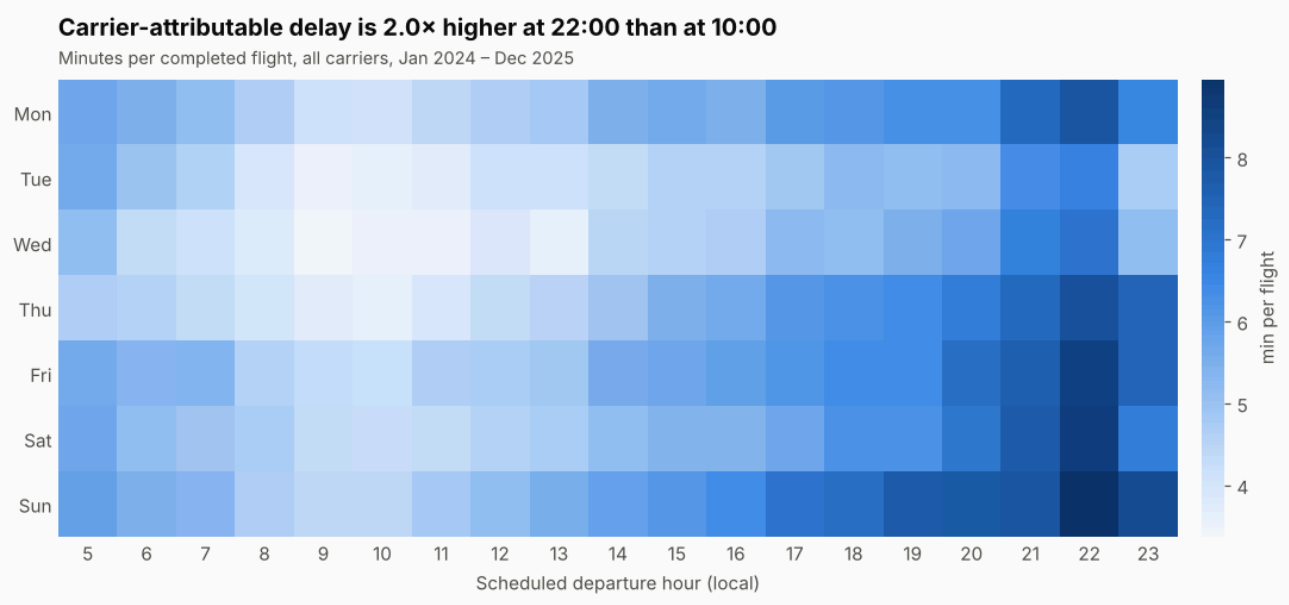
Finding 2 · Airports. 3 of 102 major airports meet every test:

- **Detroit, MI (DTW)**: 7.5 min per departure (1.44×; 1.25× against the same airlines elsewhere). 250,604 departures; main operators Delta (DL) 47%, SkyWest (OO) 21%.
- **Madison, WI (MSN)**: 7.3 min per departure (1.40×; 1.26× against the same airlines elsewhere). 24,110 departures; main operators SkyWest (OO) 21%, American (AA) 17%.
- **Des Moines, IA (DSM)**: 6.7 min per departure (1.29×; 1.27× against the same airlines elsewhere). 29,443 departures; main operators American (AA) 15%, United (UA) 14%.

Airports matter less than carriers. The airport effect is smaller once airline mix is controlled for, and the flagged airports add up to 5,600 excess hours a year, against 68,100 for the flagged carriers.

3 · Time of day and day of week

By **scheduled** departure hour (local time), so a late flight doesn't move into a later hour.



Finding 3 · Time of day and day of week. Across the network, carrier-attributable delay is lowest in the late morning (3.9 min per flight at 10:00) and climbs through the afternoon and evening to 8.0 min at 22:00. Sunday (6.0 min) is the worst day and Wednesday (4.5) the best. The weekday effect is smaller than the hour-of-day effect.

For the flagged carriers, the gap over the network is widest early in the day, when the network itself runs close to its best. Before 09:00 they run 1.76× and 1.51× the network rate. That band is 21% and 26% of their flights but 28% and 33% of their excess delay hours. Most of these flights start the day from an overnight position. That points at readiness (maintenance sign-off, crew, aircraft preparation), though the BTS data can't say which.

Each carrier's rate against the network in each time band, and the excess delay hours a year:

Scheduled departure	SkyWest (OO)	American (AA)
<09:00	1.76× · 10,300 h/yr	1.51× · 10,300 h/yr
09–13	1.62× · 8,700 h/yr	1.33× · 4,900 h/yr
13–17	1.48× · 8,100 h/yr	1.29× · 5,100 h/yr
17–21	1.46× · 8,400 h/yr	1.33× · 7,100 h/yr
21+	1.36× · 1,700 h/yr	1.40× · 3,600 h/yr

Recommendation

Fix the first wave at SkyWest (OO) and American (AA). Their pre-09:00 departures run 1.8× and 1.5× the network's carrier-attributable delay rate. Bringing them to the network rate would remove about **20,600 delay hours a year** (SkyWest (OO) 10,300, American (AA) 10,300). Likely places to look, which BTS can't separate: overnight maintenance sign-off, crew show-up and aircraft readiness (cleaning, catering, fuelling) at the stations that launch the first wave. All of these are inside the airline's control, which is what this metric measures.

The number is conservative. A late first departure pushes delay onto later flights, and BTS books that as *late-arriving aircraft* delay (40% of all delay minutes), which is not counted here.

Caveats

- **Carrier delay ≠ mechanical delay.** BTS records maintenance in the same field as crew, cleaning, fuelling and baggage.
- **Short delays aren't counted.** Delay causes are only recorded for flights 15+ minutes late.
- **Knock-on delay is left out.** Late-arriving-aircraft delay is partly the carrier's own knock-on delay, but BTS doesn't record what caused it. Leaving it out of "carrier-attributable" makes the estimate conservative.
- **SkyWest is a regional carrier.** It flies under the United, Delta, American and Alaska brands but operates and reports these flights itself. Its delays are its own operation's, not the brand's.
- **The CrowdStrike outage doesn't change the main finding.** The IT outage of 19 July 2024 hit Delta hardest. With 19–28 July 2024 removed, SkyWest and American stay flagged (see appendix). Frontier, just under the bar in 2024 (1.17×), also crosses it (1.22×), because the outage week pushed the network average up.
- **Delay causes are self-reported** by airlines. BTS covers US domestic flights by carriers above a revenue threshold, so the results may not carry over to other markets such as Changi.

Future work (out of scope for this pass)

- Weather delay by airport and season
- Route and distance effects
- Station-level first-wave analysis for the flagged carriers (which origin airports drive the pre-09:00 excess)
- Predicting carrier-attributable delay; an interactive dashboard (Streamlit)

Appendix: supporting tables

A1 · Delay minutes by cause, all airlines

Cause	Share of delay minutes	Hours a year
Late-arriving aircraft	40%	715,305
Carrier	33%	601,247
NAS (air traffic system)	20%	367,548
Weather	6%	110,872
Security	0%	2,773

Causes are recorded for flights arriving 15+ minutes late.

A2 · Airlines

Airline	Flights	Min per flight	Flights with carrier delay	× network	× others at same airports	× network 2024	× network 2025	Excess hours a year
SkyWest (regional) · OO	**1,559,088**	**8.0**	**11%**	**1.54×**	**1.34×**	**1.51×**	**1.56×**	**36,544**
American · AA	**1,918,957**	**7.2**	**15%**	**1.38×**	**1.33×**	**1.43×**	**1.32×**	**31,553**
JetBlue · B6	462,386	6.7	17%	1.28×	1.26×	1.54×	1.02×	5,652
Allegiant · G4	244,774	6.5	11%	1.24×	1.08×	1.31×	1.19×	2,603
Frontier · F9	397,672	6.3	13%	1.20×	1.22×	1.17×	1.24×	3,498
PSA (regional) · OH	460,355	6.2	12%	1.19×	1.11×	1.01×	1.36×	3,894
Delta · DL	2,010,253	6.1	10%	1.16×	1.08×	1.15×	1.18×	14,112
Hawaiian · HA	156,968	4.9	14%	0.95×	1.47×	0.81×	1.07×	-371
Spirit · NK	446,824	4.3	11%	0.82×	0.83×	0.87×	0.76×	-3,519
United · UA	1,532,195	4.1	11%	0.79×	0.81×	0.83×	0.76×	-13,987
Endeavor (regional) · 9E	195,072	3.9	7%	0.75×	0.67×	0.76×		-2,129
Alaska · AS	482,250	3.3	8%	0.62×	0.75×	0.63×	0.62×	-7,856
Southwest · WN	2,781,434	3.1	12%	0.60×	0.71×	0.60×	0.61×	-47,927
Republic (regional) · YX	629,350	3.1	7%	0.59×	0.61×	0.49×	0.68×	-11,109
Envoy (regional) · MQ	567,152	2.9	9%	0.56×	0.50×	0.60×	0.51×	-10,958

Bold = flagged: at least 1.2× in every × column. Airlines with at least 50,000 completed flights.

A3 · Airlines with the CrowdStrike outage window (19–28 July 2024) removed

Airline	Min per flight	× network	× others at same airports	× network 2024	× network 2025
SkyWest (regional) · OO	**8.0**	**1.57×**	**1.35×**	**1.56×**	**1.56×**
American · AA	**7.1**	**1.39×**	**1.34×**	**1.45×**	**1.32×**
JetBlue · B6	6.6	1.30×	1.28×	1.60×	1.02×
Allegiant · G4	6.4	1.26×	1.09×	1.34×	1.19×
Frontier · F9	**6.3**	**1.23×**	**1.24×**	**1.22×**	**1.24×**
PSA (regional) · OH	6.1	1.20×	1.12×	1.01×	1.36×
Delta · DL	5.7	1.11×	1.05×	1.05×	1.18×
Hawaiian · HA	4.9	0.97×	1.48×	0.85×	1.07×
Spirit · NK	4.1	0.81×	0.82×	0.86×	0.76×
United · UA	4.0	0.78×	0.80×	0.81×	0.76×
Endeavor (regional) · 9E	3.7	0.72×	0.64×	0.74×	
Alaska · AS	3.2	0.64×	0.76×	0.65×	0.62×
Southwest · WN	3.1	0.61×	0.72×	0.61×	0.61×
Republic (regional) · YX	3.0	0.59×	0.61×	0.48×	0.68×
Envoy (regional) · MQ	2.9	0.56×	0.50×	0.62×	0.51×

Bold = flagged. SkyWest and American stay flagged; Frontier also crosses the bar because the outage week pushed the network average up.

A4 · The 20 airports with the most carrier-attributable delay per departure

Airport	Departures	Min per departure	× network	× same airlines elsewhere	× network 2024	× network 2025	Excess hours a year
Detroit, MI · DTW	**250,604**	**7.5**	**1.44×**	**1.25×**	**1.38×**	**1.50×**	**4,757**
Madison, WI · MSN	**24,110**	**7.3**	**1.40×**	**1.26×**	**1.43×**	**1.38×**	**423**
San Juan, PR · SJU	70,036	6.8	1.30×	1.19×	1.54×	1.07×	917
Des Moines, IA · DSM	**29,443**	**6.7**	**1.29×**	**1.27×**	**1.38×**	**1.20×**	**370**
Minneapolis, MN · MSP	237,065	6.5	1.25×	1.07×	1.20×	1.31×	2,618
Miami, FL · MIA	217,835	6.5	1.24×	1.10×	1.33×	1.16×	2,308
Wichita, KS · ICT	20,995	6.5	1.24×	1.16×	1.05×	1.41×	219
Knoxville, TN · TYS	28,936	6.4	1.23×	1.09×	1.05×	1.44×	286
Atlanta, GA · ATL	646,726	6.4	1.22×	1.12×	1.15×	1.30×	6,151
Dallas/Fort Worth, TX · DFW	613,057	6.3	1.20×	1.04×	1.20×	1.20×	5,353
New York, NY · JFK	223,141	6.2	1.19×	1.08×	1.27×	1.11×	1,887
Valparaiso, FL · VPS	20,261	6.2	1.19×	1.09×	1.29×	1.10×	169
Asheville, NC · AVL	21,692	6.2	1.19×	1.08×	1.22×	1.16×	179
Albany, NY · ALB	24,169	6.1	1.18×	1.20×	1.10×	1.25×	186
Fresno, CA · FAT	24,883	6.1	1.17×	0.89×	1.24×	1.11×	182
Houston, TX · IAH	228,651	6.1	1.17×	1.23×	1.22×	1.12×	1,667
Milwaukee, WI · MKE	55,393	6.1	1.17×	1.18×	1.05×	1.28×	403
Grand Rapids, MI · GRR	36,210	6.0	1.16×	1.10×	1.06×	1.25×	244
Colorado Springs, CO · COS	25,624	5.9	1.14×	1.04×	1.02×	1.25×	155
Philadelphia, PA · PHL	193,921	5.9	1.12×	0.97×	1.08×	1.17×	1,052

Bold = flagged. Airports with at least 20,000 departures; 102 qualify.

A5 · All airlines by scheduled departure hour

Scheduled departure	Flights	Min per flight	Flights with carrier delay	× network
05:00	392,090	5.5	6%	1.05×
06:00	962,364	5.0	6%	0.97×
07:00	988,059	4.8	7%	0.93×
08:00	954,934	4.3	8%	0.83×
09:00	781,277	4.0	8%	0.76×
10:00	860,819	3.9	9%	0.76×
11:00	868,224	4.2	10%	0.80×
12:00	821,886	4.5	11%	0.86×
13:00	827,952	4.6	12%	0.88×
14:00	797,477	5.1	13%	0.98×
15:00	796,118	5.4	14%	1.03×
16:00	796,908	5.5	15%	1.05×
17:00	867,902	5.9	15%	1.13×
18:00	835,674	6.1	15%	1.17×
19:00	753,475	6.3	16%	1.21×
20:00	615,005	6.6	17%	1.27×

Scheduled departure	Flights	Min per flight	Flights with carrier delay	× network
21:00	459,680	7.3	16%	1.40×
22:00	315,814	8.0	16%	1.54×
23:00	113,064	6.7	13%	1.28×

00:00–04:59 is left out: 36,008 flights, under 0.3% of the total.

A6 · All airlines by day of week

Day	Flights	Min per flight	Flights with carrier delay	× network
Monday	2,088,250	5.4	12%	1.03×
Tuesday	1,893,059	4.5	9%	0.87×
Wednesday	1,910,400	4.5	10%	0.86×
Thursday	2,046,133	5.1	12%	0.98×
Friday	2,061,923	5.6	12%	1.07×
Saturday	1,788,896	5.3	12%	1.02×
Sunday	2,056,069	6.0	13%	1.15×

A7 · Flagged airlines by time of day

Airline	Scheduled departure	Flights	Min per flight	Network min per flight	× network	Excess hours a year	Share of flights	Share of excess
SkyWest (regional) · OO	<09:00	332,825	8.6	4.9	1.76×	10,266	21%	28%
SkyWest (regional) · OO	09–13	403,161	6.7	4.1	1.62×	8,679	26%	23%
SkyWest (regional) · OO	13–17	396,623	7.6	5.1	1.48×	8,079	25%	22%
SkyWest (regional) · OO	17–21	350,358	9.1	6.2	1.46×	8,389	22%	23%
SkyWest (regional) · OO	21+	76,121	10.2	7.5	1.36×	1,693	5%	5%
American · AA	<09:00	505,932	7.3	4.9	1.51×	10,349	26%	33%
American · AA	09–13	428,369	5.5	4.1	1.33×	4,856	22%	16%
American · AA	13–17	420,656	6.6	5.1	1.29×	5,133	22%	17%
American · AA	17–21	418,669	8.2	6.2	1.33×	7,065	22%	23%
American · AA	21+	145,331	10.5	7.5	1.40×	3,603	8%	12%

Flights are totals for the whole period; excess hours are per year.